

Proportional Relationships

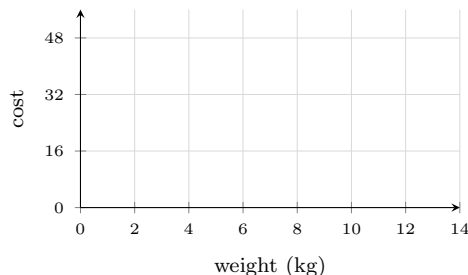
Moving among tables, graphs, equations, and contexts

Directions

- In a proportional relationship $y = kx$, the number k is the *constant of proportionality* — the value of y when x is 1.
- Every problem asks you to move between representations. Show the equation you are using, even when the question starts from a table or a graph.
- Work in the space on the left of each problem; use the grid on the right to plot or sketch the relationship.

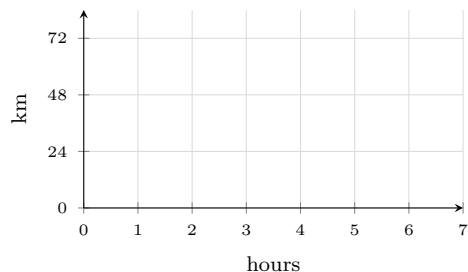
1. The table shows what oranges cost at a market. (a) What do 8 kg of oranges cost? _____ (b) Find the constant of proportionality (the cost of one kilogram). _____ (c) Write the equation linking cost c to weight w . _____ (d) Plot the four pairs on the grid.

kg	2	5	8	12
dollars	8	20	32	48

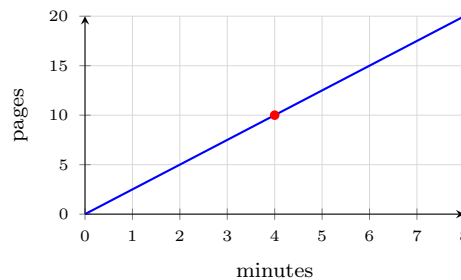


2. The table shows how far a cyclist travels at a steady speed. (a) How far does the cyclist travel in 4 hours? _____ (b) Find the constant of proportionality. _____ (c) Use your equation to find the distance travelled in 10 hours. _____ (d) Plot the four pairs on the grid.

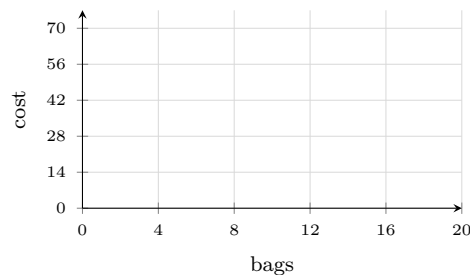
hours	1	3	4	6
km	12	36	48	72



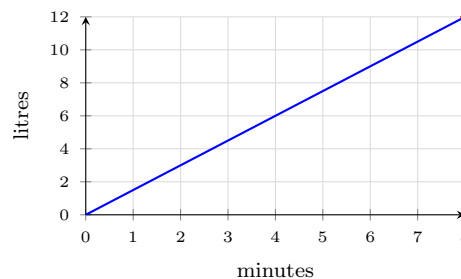
3. The graph shows how many pages Amir has read after x minutes. Find the unit rate, in pages per minute, _____ and write the equation of the line. _____



4. A shop charges a proportional amount for bags of soil: the equation $c = 3.5n$ gives the cost c in dollars for n bags. (a) Find the cost of 12 bags. _____ (b) A gardener spends exactly 70 dollars. How many bags is that? _____ (c) Sketch the graph of $c = 3.5n$ on the grid.

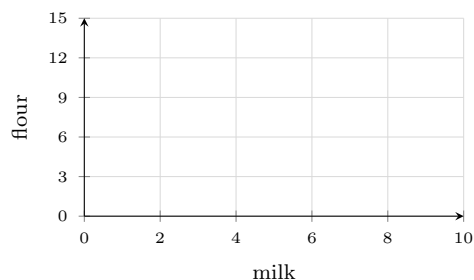


5. The graph shows the water in a tank as it fills. (a) Find the unit rate, in litres per minute. _____ (b) Use the equation to find how much water is in the tank after 10 minutes.



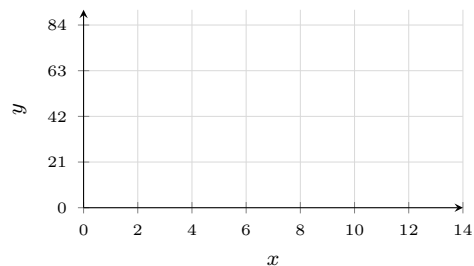
Answer: _____ L

- 6.** A bread recipe uses 3 cups of flour for every 2 cups of milk, and the amounts stay in that ratio. (a) Find the constant of proportionality for cups of flour f per cup of milk m . _____ (b) Write the equation. _____ (c) How much flour goes with 9 cups of milk? _____ (d) Sketch the relationship on the grid.



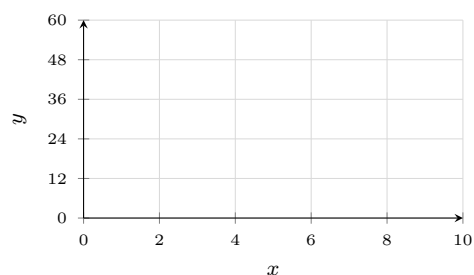
- 7.** This table shows a proportional relationship, but one value is missing. (a) Find the constant of proportionality. _____ (b) Find the missing x -value. _____ (c) Plot the three complete pairs on the grid.

x	3	5	?	12
y	21	35	56	84

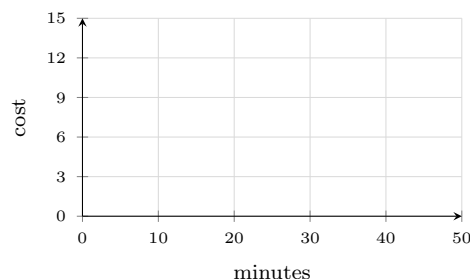


- 8.** The equation $y = 6x$ describes a proportional relationship. (a) Complete the table. (b) Plot your three pairs on the grid and draw the line through them and the origin.

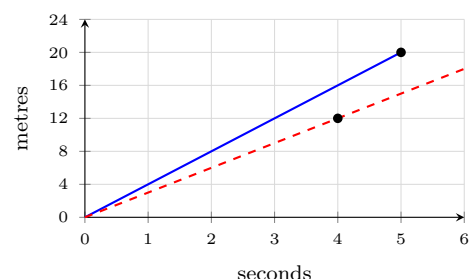
x	2	5	9
y			



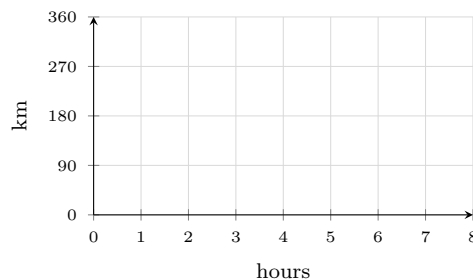
- 9.** Two phone plans. Plan A charges 0.25 dollars for every minute of calls. Plan B charges 5 dollars a month plus 0.15 dollars for every minute. (a) What does a 40 minute month cost on Plan A? _____ (b) On Plan B, find the cost per minute for a 40 minute month _____ and then the cost per minute for a 20 minute month. _____ (c) What do your two answers in part (b) show about whether Plan B is a proportional relationship? Write a sentence in the space at the left. (d) Sketch both plans on the grid.



- 10.** The graph shows the distance run by two runners. Runner *A* is the solid line and runner *B* is the dashed line. (a) Find each runner's unit rate, in metres per second. Runner *A*: _____ Runner *B*: _____ (b) Write $<$, $>$ or $=$ between runner *A*'s rate and runner *B*'s rate. _____



- 11.** A car travels at a steady speed, modelled by $d = 45t$, where d is the distance in kilometres after t hours. (a) How far does the car travel in 2.5 hours? (b) How many hours does it take to travel 315 km? (c) Sketch the graph on the grid.



Answer: _____ km

Answer: _____ hours

12. A print shop charges a proportional amount for posters. Two past orders are recorded in the table. (a) Find the constant of proportionality from the second order. _____ (b) Write the equation. _____ (c) What would 35 posters cost? _____ (d) A customer paid 102 dollars — how many posters was that? _____ (e) Plot both orders on the grid and draw the line through them and the origin.

posters	8	20
dollars	24	60

